Paleomagnetic Comparison of Cretaceous Rocks from South Korea and Late Paleozoic and Mesozoic Rocks of Japan

From five widely spaced areas in South Korea 35 oriented rock samples were collected for the determination of their natural remanent magnetization (NRM) properties from Cretaceous andesites, dacites, and rhyolites. Magnetic cleaning in a maximum ac field of 400 œ. was employed on 23 of these samples to decrease scattering. The mean declinations and inclinations of the NRM and the 95% cones of confidence are given in Table I.

Table I
DIRECTIONS AND INTENSITIES OF MAGNETIZATION FROM CRETACEOUS ROCKS IN SOUTH KOREA

Location	N	Lat.	Long.	Rock Type	D	I	K	α 95%	$J_r \times 10^3$
Pohang (2)	6	36.2N	129.3E	rhyolite	18.5	59.8	42.3	8.8	0.025
Shin-Lee (1)*	8	35.9N	127.5E	andesite	14.6	46.2	39.6	10.0	6.670
Gyongsan (3)*	6	35.8N	128.7E	andesite	21.4	39.4	74.4	6.6	0.125
Anjong-dong (4)*	9	35.4N	128.9	dacite	42.5	62.0	37.5	11.4	9.200
Kimhae (5)	6	35.2N	128.8	andesite	14.9	54.0	18.5	13.3	0.260
Means					19.5	53.3	43.6	9.48	

* Represent only demagnetized results.

In most cases the rocks contained both magnetite and hematite with amounts of each ranging from traces to a maximum of 2.8% magnetite at Gyongson and 4.6% hematite at Kimhae. Ilmenite of only 0.2% was observed from the rocks of the Kimhae area. No local or major structural features were observed in the field thus none of the samples were subjected to tilt corrections. All of the rocks with the exception of the rhyolite at Pohang were fresh and showed no weathering or alteration in situ. Results from Pohang and Kimhae, although rather weak in magnetic intensity, gave good groupings with their directions of magnetization similar to those directions of the demagnetized samples.

The directions of magnetization for the Cretaceous rocks from South Korea are similar to those for late Paleozoic and Mesozoic rocks from southwest Japan but are markedly different than the directions of magnetization for Mesozoic rocks from northeast Japan (Fig. 1). Kawai, et al (1961, 1962) have attributed these differences in declination from the two sectors of the islands to deformation in early Tertiary time and that the bending of the Japanese Islands took place along a zone of plastic deformation referred to as the "fossa magna". They inferred that the amount of relative rotation between these two sections of the islands to be about 40 degreees.

In an attempt to compare differential movement of the Japanese Islands with South Korea the mean virtual geomagnetic pole was calculated for the South Korea Cretaceous rocks. This mean pole position, along with VGP's for the southwestern and northeastern

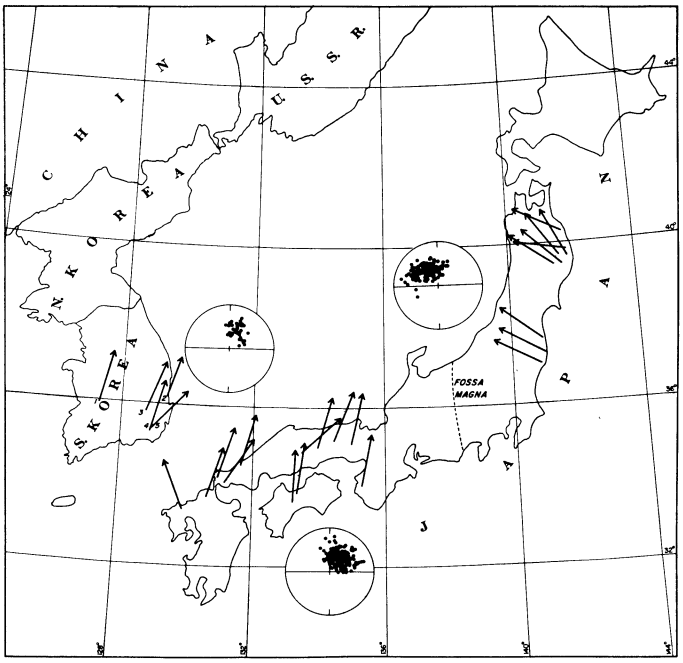


Fig. 1 Directions of magnetization for Paleozoic and Mesozoic rocks from Japan (Kawai, *et. al.*) and Cretaceous rocks from South Korea.

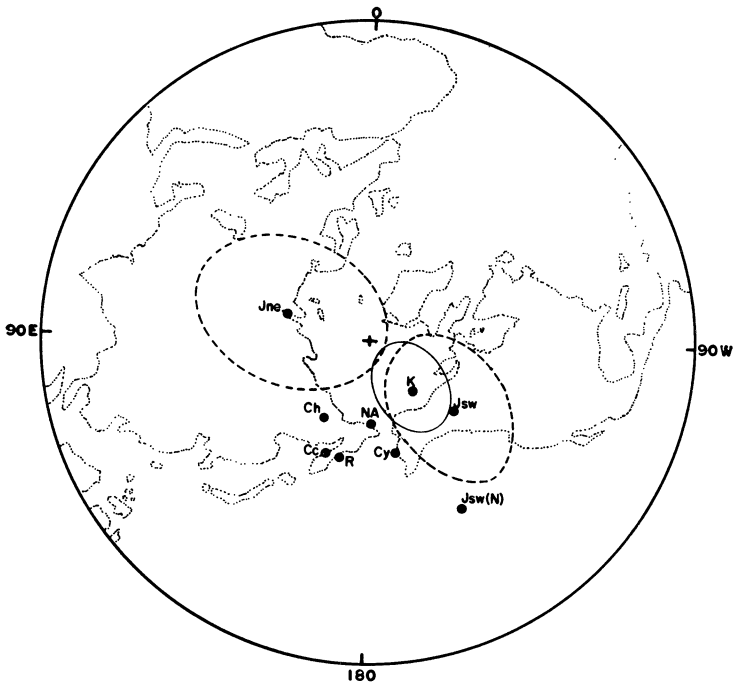


Fig. 2 Mean Virtual Geomagnetic Poles for Cretaceous, Mesozoic and Cenozoic rocks from Southwest Japan (Jsw), Northeastern Japan (Jne), South Korea (K), North America (NA), Russia, (R), and China (Ch, Cc, and Cy),

Table II
MEAN VIRTUAL GEOMAGNETIC POLES FOR KOREA, JAPAN, NORTH AMERICA, CHINA, AND RUSSIA

Location	Lat.	Long.	Geological Period	VGP Lat.	VGP Long.	p	m
South Korea			Cretaceous	73.9N	140.6W	9.16	13.18
Southwest Japan (Irving)			Late Paleozoic and	63.0N	130.0W	14.0	21.0
(Nagata)			Mesozoic	42.0N	153.3W	2.0	3.0
Northeast Japan (Irving)			Mesozoic	68.0N	73.0E	20.0	25.0
North America			Mesozoic	69.0N	177.0W		
China							
(Ch) Huipei	29.3N	111.2E	Mesozoic and	68 N	151 E		
(Cc) Canton	23.2N	114.2E	Cenozoic	60.1N	159.0E	6.65	8.69
(Cy) Yuannan	25 N	102.1E		60 N	174.0W		
Russia			Mesozoic	60 N	167 E		

Japanese Island segments, (Irving, 1964), northern part of North America, (Irving, 1964), eastern Russia, (Lee, 1963), and south and central China, (Lee, 1963) are listed in Table II and plotted on Figure 2.

Based on the VGP values, southwest Japan has rotated about 15 degrees clockwise and northeast Japan has rotated about 40 degrees counter-clockwise relative to South Korea. This is in agreement with Kawai and Abe (1963) who, using data "obtained at various parts of the world," estimated that the rotations were 10 and 40 degrees respectively for southwest and northeast Japan. This would indicate that South Korea, relative to Japan has experienced little or no horizontal angular movement since Cretaceous time.

It is known that certain geological events have occurred since early Mesozoic time in South Korea (Gallagher and Chong, 1946). At the end of the Jurassic and the beginning of early Cretaceous a pronounced orogeny occurred producing tectonic directions trending north-northeasterly in South Korea. At the end of the Cretaceous faulting along with igneous activity took place. Many directions of faulting are evident for this time but the most pronounced are those trending north-northwesterly. Some attribute these faults as having been the major cause of the present southward projection of the Korean penninsula from the Asiatic mainland. Further the east coast of Korea is a smooth, steep, shore line of emergence while the western and southern coasts exhibit shore lines of submergence.

Correcting the VGP positions for the indicated degrees of rotation from southwest and northeast Japan the new VGP positions would be very close to that of the Cretaceous pole of South Korea. It is noted that the new positions for Japan's VGP's and that of South Korea would not group with those obtained from North America and the Asiatic mainland. Not knowing the accuracy, reliability or 95% cones of confidence of these latter data it becomes prohibitive to subject Japan and South Korea either individually or collectively to any horizontal and/or vertical motions to bring their VGP's into the area of grouping of the other Cretaceous pole positions as plotted.

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